A mass gap and conserved-momentum geometry for negative-energy biharmonic traveling waves

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Abstract

For the focusing mass-critical biharmonic nonlinear Schrödinger equation, we prove a positive gap between the sharp Gagliardo–Nirenberg threshold mass and the mass of any negative-energy traveling profile, in every spatial dimension. The proof uses compactness of optimizing sequences and an elementary rigidity argument for the phase Hessian. It requires neither uniqueness nor real-part nondegeneracy of ground states, and no numerical profile certificate. We also give an exact positive factorization of energy along the entire dilation curve preserving both mass and momentum of a negative-energy wave. These results identify two structural restrictions on coherent escape from collapse. They do not classify all globally bounded solutions.

The proofs below are self-contained apart from standard Sobolev, multiplier and compactness facts, whose applications are specified. The main theorems are independent of the computational certificates and multiwave constructions in the earlier project.

1 Equation and main mass-gap statement

Fix $N \geq 1$ and write

$$\alpha = \frac{8}{N}, \qquad p = 2 + \alpha, \qquad r = \frac{4}{N}, \qquad p = 2(r+1).$$

The equation and profile equation are

$$\mathrm{i}u_t = \Delta^2 u - |u|^\alpha u, \tag{1}$$

$$\Delta^2 U + \omega U + \mathrm{i}v \cdot \nabla U - |U|^\alpha U = 0, \qquad u(t,x) = e^{\mathrm{i}\omega t} U(x - vt). \tag{2}$$

Here $\omega \in \mathbb{R}$, $v \in \mathbb{R}^N$, and a profile is a nonzero weak solution $U \in H^2(\mathbb{R}^N; \mathbb{C})$. Set

$$M(f) = \|f\|_2^2, \quad A(f) = \|\Delta f\|_2^2, \quad E(f) = \frac{1}{2}A(f) - \frac{1}{p}\|f\|_p^p, \quad P(f) = \operatorname{Im}\int \bar{f}\nabla f.$$

Let $M_c > 0$ be defined by the sharp inequality

$$\|f\|_p^p \leq \frac{p}{2}\left(\frac{M(f)}{M_c}\right)^r A(f). \tag{3}$$

Equivalently, M_c is the mass of a ground state with nonlinear coefficient one. Optimizers can be normalized to have $M = M_c$ and $A = 1$.

Theorem 1.1 (A uniform gap for all traveling profiles)**.** *For every $N \geq 1$ there is $\varepsilon_N > 0$ such that every profile of (2) with $E(U) < 0$ satisfies*

$$\boxed{M(U) \geq M_c + \varepsilon_N.}$$

The gap is uniform in frequency, velocity, translations and phase. No radial or parity assumption is imposed, and no finite-variance hypothesis is needed. The proof is qualitative: it does not supply a numerical value of ε_N.

The theorem is an exclusion result; it does not assert existence of negative-energy waves in every dimension. It is stronger than the immediate consequence $M(U) > M_c$ of (3), since a sequence of strictly larger masses could otherwise converge to M_c.

2 Profile identities at energy regularity

Every weak profile in (2) belongs to H^4. Here are sufficient details to handle small, noninteger α in large dimension. For $N \le 8$, Sobolev embedding gives $U \in L^{2(1+\alpha)}$ directly, so $|U|^\alpha U \in L^2$ and (2) gives H^4 regularity. For $N > 8$, choose μ so large that

$$|\xi|^4 - v \cdot \xi + \mu \ge c(1 + |\xi|^4).$$

The inverse $R_\mu = (\Delta^2 + \mathrm{i}v \cdot \nabla + \mu)^{-1}$ maps L^s to $W^{4,s}$ for $1 < s < \infty$, by the multiplier theorem. Write

$$U = R_\mu(|U|^\alpha U) + (\mu - \omega)R_\mu U.$$

The second term is in H^4, hence in $L^{q_{\mathrm{lin}}}$, $q_{\mathrm{lin}} = 2N/(N-8)$. Starting from $q_0 = 2N/(N-4)$, the first term improves reciprocal integrability according to

$$a_{j+1} = (1 + 8/N)a_j - 4/N, \qquad a_0 = \tfrac{1}{2} - \tfrac{2}{N},$$

until $q = 2(1+\alpha)$ is reached. The ideal iterates are $a_j = \frac{1}{2} - (2/N)(1+8/N)^j$, so this takes finitely many steps. Intermediate exponents may be slightly weakened to avoid endpoints. All input exponents $q_j/(1+\alpha)$ exceed one, and $2(1+\alpha) < q_{\mathrm{lin}}$. Thus the linear term does not obstruct the iteration. Now the nonlinearity is in L^2, and the original equation implies $U \in H^4$.

Lemma 2.1 (Pohozaev and current identities)**.** *Every such profile satisfies, with the unitary Fourier transform,*

$$v \cdot P(U) = 8E(U), \tag{4}$$

$$W(U) := \int |\xi|^2 \xi |\widehat{U}(\xi)|^2 \, \mathrm{d}\xi = \frac{M(U)}{4} v, \tag{5}$$

$$\omega M(U) = rA(U) + (8-p)E(U). \tag{6}$$

For $U = a + \mathrm{i}b$, with real a, b, put

$$\mathcal{A}_U = \Delta^2 + \omega - |U|^\alpha.$$

Then, in quadratic-form notation,

$$\boxed{E(U) = \tfrac{1}{4} \langle b, \mathcal{A}_U b \rangle \, .} \tag{7}$$

Proof. The profile is a critical point of $S = E + \omega M/2 - v \cdot P/2$. Mass-preserving dilation scales E and P by λ^4 and λ, respectively, giving (4). Linear phase modulation $e^{\mathrm{i}k \cdot x}U$ has energy derivative $2W$ and momentum derivative M at $k = 0$, giving (5).

These variations do not require weighted assumptions. To justify them at H^4, test the weak equation with the cutoff dilation

$$(x\chi(x/R)) \cdot \nabla U + \tfrac{1}{2} \operatorname{div}(x\chi(x/R))U$$

and the cutoff phase $ix_j\chi(x/R)U$, respectively. Here χ is smooth, compactly supported and one near the origin. The tests belong to H^2. Integration by parts gives the displayed identities in the limit $R \to \infty$: coefficients of the leading terms converge to their constant values, and the additional derivatives of the cutoff are $O(R^{-1})$ or smaller. Their terms are bounded by tails of $|U|^2$, $|\nabla U|^2$, $|\Delta U|^2$, and $|U|^p$, or products controlled by these norms. All tails vanish. The nonlinear integration by parts is valid since $|U|^\alpha U \in L^2$ and $\nabla U \in L^2$.

Taking the real pairing with U gives $A + \omega M - v \cdot P - \|U\|_p^p = 0$; combine this with $\|U\|_p^p = (p/2)A - pE$ to obtain (6). Finally, the imaginary profile equation is $\mathcal{A}_U b = -v \cdot \nabla a$. Since $P(U) = -2\int b\nabla a$, it gives $v \cdot P = 2\langle b, \mathcal{A}_U b\rangle$, proving (7). □

3 Phase rigidity without uniqueness or radial symmetry

The real-valuedness argument below is a standard strict-convexity mechanism; compare the phase-rotation argument in Proposition 3.8 of [1]. Fourier rearrangement provides further radial symmetry for even powers [2], but radial symmetry is not needed here.

Lemma 3.1 (All optimizers have constant phase)**.** *Every complex optimizer of* (3) *is a constant phase times a real function.*

Proof. Let $f = a + ib$. With $C = (p/2)M_c^{-r}$, Minkowski, (3) on the real components, and two-term Hölder give

$$\begin{aligned}\|f\|_p^2 &\le \|a\|_p^2 + \|b\|_p^2 \\ &\le C^{2/p}\big[A(a)^{2/p}M(a)^{1-2/p} + A(b)^{2/p}M(b)^{1-2/p}\big] \\ &\le C^{2/p}A(f)^{2/p}M(f)^{1-2/p}.\end{aligned}$$

For an optimizer every inequality is an equality. Strict convexity of $L^{p/2}$ gives $b^2 = ca^2$ almost everywhere, unless one component vanishes. Apply the same equality argument to $e^{i\pi/4}f$. If both signs of $b/a = \pm\sqrt{c}$ occurred on sets of positive measure, the new two squared components could not be proportional. For $c = 1$ they instead have disjoint nonzero supports, which also gives strict Minkowski inequality. Hence one sign holds almost everywhere, proving the claim. The same argument covers any initial constant phase rotation. □

Let Q be a real optimizer normalized by $M(Q) = M_c$ and $A(Q) = 1$. Its Euler–Lagrange equation is

$$\Delta^2 Q + \omega_Q Q - |Q|^{2r}Q = 0, \qquad \omega_Q = r/M_c > 0.$$

Lemma 3.2 (Strict positivity after removing phase)**.** *For every such optimizer,*

$$L_- := \Delta^2 + \omega_Q - |Q|^{2r} \ge 0, \qquad \ker L_- = \operatorname{span}\{Q\}.$$

There is $c_Q > 0$ such that for every real $b \in H^2$ with $\langle b, Q\rangle = 0$,

$$\langle b, L_- b\rangle \ge c_Q \|b\|_{H^2}^2 .$$

Proof. We retain $A = A(Q)$ and $M = M(Q)$ in the algebra; $\omega_Q = rA/M$. For real b, set $B = M(b)$, $C_1 = A(b)$ and

$$F(s) = \int (Q^2 + sb^2)^{r+1}, \qquad s \ge 0.$$

The sharp inequality and equality at Q imply

$$F(s) \le (r+1)(1 + sB/M)^r(A + sC_1). \tag{8}$$

Comparison of right derivatives at zero gives $L_- \geq 0$. If b belongs to its form kernel, these first derivatives agree and

$$\int |Q|^{2r} b^2 = C_1 + rAB/M.$$

Compare the second-order remainders in (8). Convexity gives a nonnegative pointwise remainder. Fatou's lemma, or the integral Taylor formula, yields

$$D := \int |Q|^{2r-2} b^4 \leq (r-1)AB^2/M^2 + 2BC_1/M. \tag{9}$$

Here the integrand at $Q = 0$ is interpreted by the limiting second-order remainder: it is infinite if $r < 1$ and $b \neq 0$, equals b^4 if $r = 1$, and is zero if $r > 1$. Thus when $r < 1$ the finite upper bound already forces $b = 0$ almost everywhere on that zero set.

Cauchy–Schwarz gives

$$D \geq \frac{(C_1 + rAB/M)^2}{(r+1)A}.$$

Combining this with (9), and multiplying by $(r+1)A$, gives

$$(C_1 - AB/M)^2 \leq 0.$$

Thus all inequalities are equalities. The Cauchy equality gives $b^2 = (B/M)Q^2$ on $\{Q \neq 0\}$. Its integral already equals B, so $b = 0$ on the remaining zero set as well, including the case $r > 1$. Moreover $C_1 = AB/M$.

The form kernel is a linear space containing Q. Applying the same conclusion to $b + Q$ yields $(b+Q)^2 = c'Q^2$. Subtracting $b^2 = (B/M)Q^2$ gives $bQ = c''Q^2$; since b already vanishes where Q does, $b = c''Q$ almost everywhere. This proves the kernel statement without a maximum principle or a Fourier-positivity argument.

Finally, multiplication by $|Q|^{2r}$ is compact from H^2 to H^{-2}. Indeed $|Q|^{2r} \in L^{p/(2r)}$, the exponent p is strictly below the H^2 critical embedding exponent, local Rellich compactness applies, and the tail is small by Hölder. Consequently L_- is a compact form perturbation of the positive operator $\Delta^2 + \omega_Q$. Nonnegativity and the one-dimensional kernel imply the claimed H^2 coercivity on the L^2 orthogonal complement. □

4 Compactness and proof of the uniform mass gap

Lemma 4.1 (Normalized near-optimizers are compact)**.** *Suppose $M(f_n) \to M_c$, $A(f_n) = 1$, and $E(f_n) < 0$. After a subsequence and translations, f_n converges strongly in H^2 to a normalized optimizer.*

Proof. The sharp inequality sandwiches the nonlinear norm:

$$\frac{p}{2} < \|f_n\|_p^p \leq \frac{p}{2}\left(\frac{M(f_n)}{M_c}\right)^r \longrightarrow \frac{p}{2}.$$

In particular $E(f_n) \to 0$. Nonvanishing for bounded H^2 sequences follows from the local Sobolev inequality and a covering by unit cubes: if the mass on every translated fixed cube tended to zero, the L^p norm would tend to zero. Thus translations yield a nonzero weak limit Q, also with local strong convergence and almost-everywhere convergence.

Let $m_1 = M(Q) > 0$, $a_1 = A(Q) > 0$, $m_2 = M_c - m_1$ and $a_2 = 1 - a_1$. Hilbert norm splitting and the Brezis–Lieb lemma, followed by (3), give

$$M_c^r \leq m_1^r a_1 + m_2^r a_2, \qquad a_1 + a_2 = 1.$$

If $0 < m_1 < M_c$, both coefficients m_j^r are strictly less than M_c^r, a contradiction. Hence $m_1 = M_c$, and the inequality then forces $a_1 = 1$. This proves strong H^2 convergence and equality in (3). □

Proof of Theorem 1.1. If no uniform gap existed, there would be profiles U_n with negative energy and $M(U_n) \downarrow M_c$. Critical scaling

$$U(x) \mapsto \lambda^{N/2} U(\lambda x), \qquad (\omega, v) \mapsto (\lambda^4 \omega, \lambda^3 v)$$

preserves mass and the sign of energy. Normalize $A(U_n) = 1$. Lemmas 4.1 and 3.1 give, after translations and a constant phase, $U_n \to Q$ strongly in H^2, with Q real. By (6),

$$\omega_n \longrightarrow r/M_c = \omega_Q.$$

Rotate each $U_n = a_n + \mathrm{i}b_n$ by a further phase tending to zero so that $\langle b_n, Q\rangle = 0$. This is possible since $\langle U_n, Q\rangle \to M_c$.

The forms $\mathcal{A}_n = \Delta^2 + \omega_n - |U_n|^{2r}$ converge in operator norm $H^2 \to H^{-2}$ to L_-. To see this, $U_n \to Q$ in L^p implies $|U_n|^{2r} \to |Q|^{2r}$ in $L^{p/(2r)}$, and Hölder with two L^p factors bounds the form difference. Lemma 3.2 therefore gives, for large n,

$$\langle b_n, \mathcal{A}_n b_n\rangle \geq \tfrac{1}{2} c_Q \, \|b_n\|_{H^2}^2 \geq 0.$$

But (7) says $E(U_n) = \langle b_n, \mathcal{A}_n b_n\rangle / 4 < 0$. This contradiction proves the uniform gap. No uniform coercivity constant over all ground states is needed: compactness selects one limiting optimizer. □

Corollary 4.2 (Arbitrary elliptic quartic anisotropy)**.** *Let $a(\xi)$ be a real homogeneous polynomial of degree four satisfying*

$$0 < c|\xi|^4 \leq a(\xi) \leq C|\xi|^4 \qquad (\xi \neq 0).$$

Replace Δ^2 by $a(D)$ and $A(f)$ by $A_a(f) = \int a(\xi)|\widehat{f}(\xi)|^2 \,\mathrm{d}\xi$. Define E_a and the sharp threshold mass $M_{c,a}$ accordingly. There is $\varepsilon_{N,a} > 0$ such that every H^2 profile solving

$$a(D)U + \omega U + \mathrm{i}v \cdot \nabla U - |U|^{8/N} U = 0, \qquad E_a(U) < 0,$$

satisfies $M(U) \geq M_{c,a} + \varepsilon_{N,a}$.

Proof. The symbol is even, so the kinetic form is real and splits additively over real and imaginary components. Its norm is equivalent to the homogeneous H^2 norm, and its homogeneity gives the same dilation identity $v \cdot P = 8E_a$. The elliptic regularity argument applies to $a(D) + \mathrm{i}v \cdot \nabla + \mu$ for large μ. All GN, phase-kernel and compactness arguments above use only these properties. The identity $E_a(U) = \left\langle b, (a(D) + \omega - |U|^{8/N})b \right\rangle / 4$ is unchanged. The same contradiction proves the result. No smallness of the anisotropy is assumed; the gap is not asserted uniform over all symbols. □

5 Exact geometry at fixed mass and momentum

This section applies to every negative-energy profile in every dimension, independently of the mass-gap proof. For such a profile put

$$a = \frac{P(U)}{M(U)}, \quad m = M(U)|a|^4, \quad B = \int (|a|^2|\xi|^2 + 2(a \cdot \xi)^2)|\widehat{U}(\xi)|^2 \,\mathrm{d}\xi.$$

The letters a, m, B here are unrelated to the real/imaginary decomposition in earlier sections. Jensen gives $B \geq 3m$.

Theorem 5.1 (A strict minimum on the conserved-momentum dilation curve)**.** *Define*

$$U_\lambda(x) = e^{\mathrm{i}(1-\lambda)a \cdot x} \lambda^{N/2} U(\lambda x), \qquad \lambda > 0.$$

Then $M(U_\lambda) = M(U)$ and $P(U_\lambda) = P(U)$. With $E = E(U) < 0$,

$$E(U_\lambda) - E = (\lambda-1)^2\big[(B - \tfrac{3}{2}m - 3E)\lambda^2 + (m-2E)\lambda + (\tfrac{1}{2}m - E)\big]. \tag{10}$$

All three coefficients in brackets are positive. In particular,

$$E(U_\lambda) - E(U) \geq \big(|E(U)| + m/2\big)(\lambda-1)^2.$$

Thus U is the unique energy minimum on this entire curve.

Proof. For $U_{\lambda,k} = e^{ik\cdot x}\lambda^{N/2}U(\lambda x)$, direct Fourier expansion gives

$$E(U_{\lambda,k}) = \lambda^4 E + 2\lambda^3 k\cdot W + \lambda^2 \int (|k|^2|\xi|^2 + 2(k\cdot\xi)^2)|\widehat{U}|^2 + 2\lambda|k|^2 k\cdot P + \tfrac{1}{2}M|k|^4.$$

The nonlinear energy scales as λ^4. Equations (4)–(5) give $a\cdot W = 2E$. Substitute $k = (1-\lambda)a$ and factor to get (10). Positivity follows from $B \geq 3m$ and $E < 0$. □

Two useful consequences are

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2}E(U_\lambda)\bigg|_{\lambda=1} = -12E + 2B > 0,$$

and

$$E(e^{-\mathrm{i}a\cdot x}U) = B - \tfrac{3}{2}m - 3E \geq \tfrac{3}{2}m + 3|E| > 0. \tag{11}$$

So neutralizing a wave's momentum by a linear phase necessarily destroys its negative energy. Ordinary mass-preserving dilation has energy $\lambda^4 E$; it does not preserve momentum. A minimum on the particular curve above is not a minimum on the full constraint surface and does not prove orbital stability.

6 The next dynamical threshold, and a radiation constraint

Define the distinct dynamical quantity

$$m_{\mathrm{ng}} = \inf\left\{ M(u_0) : \begin{array}{l} E(u_0) < 0,\ P(u_0) = 0,\ u \text{ is forward global,} \\ \sup_{t\geq 0}\|u(t)\|_{H^2} < \infty \end{array} \right\},$$

with the infimum of the empty set equal to $+\infty$. The sharp inequality gives only $m_{\mathrm{ng}} \geq M_c$. Theorem 1.1 concerns traveling profiles; it does not yet prove $m_{\mathrm{ng}} > M_c$. Proving that stronger statement would require control of arbitrary bounded dynamics, including radiation and multiple separating cores.

For comparison, let m_{tw} be the infimum of the masses of all negative-energy traveling profiles. The theorem proves $m_{\mathrm{tw}} \geq M_c + \varepsilon_N$, with no assertion that the infimum is attained. If one already has an exact opposite-velocity two-wave solution with profile U_v, conservation and decoupling give $m_{\mathrm{ng}} \leq 2M(U_v)$. This upper bound depends on that separate nonlinear construction.

There is a useful universal constraint on a prospective wave-plus-radiation mechanism. Suppose an exact solution were asymptotic to a negative-energy traveling wave plus free radiation $e^{-\mathrm{i}t\Delta^2}f$, with vanishing cross terms and zero total momentum. For $m_f = M(f) > 0$, Jensen gives

$$E_{\mathrm{rad}} = \tfrac{1}{2}\|\Delta f\|_2^2 \geq \frac{|P(f)|^4}{2m_f^3}.$$

The constant is sharp as an infimum, approached by narrow Fourier packets. Since $P(f) = -P(U)$, strict negativity of total energy requires

$$m_f^3 > \frac{|P(U)|^4}{2|E(U)|}. \tag{12}$$

This is necessary, not a nonlinear existence theorem. For a branch with $|P(U_v)| \asymp |v|$ and $-E(U_v) \asymp v^2$, the lower bound on radiation mass from this budget is of order $|v|^{2/3}$. Thus two full nonlinear cores are not the only possible momentum-balancing mechanism. Constructing an exact radiative wave operator is a substantive next step.

Three objectives now have precise content:

1. Determine whether the traveling-wave gap extends to $m_{\mathrm{ng}} > M_c$, and identify the least-mass escape mechanism. This is a research question posed here, not an attributed named conjecture.
2. Construct an infinite-dimensional center-stable family near a negative-energy excited wave, first with uniform orbital control and then with an asymptotic description of radiation. The second objective additionally requires continuous-spectrum and internal-mode analysis.
3. Prove a local soliton-resolution theorem in a specified regime. Recent one-dimensional mixed-dispersion work explicitly formulates asymptotic stability, branch transitions and soliton-resolution conjectures [3]. Those are related targets, not consequences of the present all-dimensional mass-gap theorem, and their dimension and parameter restrictions must be preserved.

References

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